

Course: B. Tech
Semester: IV

Branch: Civil Engineering/Civil and Infrastructure Engineering/Civil and Environmental Engineering

Subject Name: Water Resources Engineering Subject Code: BTCVC404 / BTCIC404 / BTCEC404

Max Marks: 60

Duration: -3 Hrs.

Date: 20/06/2024

Instructions to the Students:

1. All questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mentioned it clearly.

	Level/CO	Marks
Q. 1 Solve any TWO of the following		12
(A) Define irrigation and discuss its necessity. Enlist the ill-effects and advantages of irrigation.	CO 1	06
(B) Derive relationship between duty and delta.	CO1	06
(C) Explain the factors affecting duty of water.	CO1	06
Q.2 Solve any TWO of the following		12
(A) State the factors governing on the selection of site for reservoir.	CO1	06
(B) Explain with neat sketch various zones in storage reservoir.	CO2	06
(C) Explain elementary profile of gravity dam.	CO2	06
Q. 3 Solve any TWO of the following		12
(A) Explain Bligh's creep theory and state its limitations.	CO2	06
(B) Explain Lacey's silt theory	CO2	06
(C) Determine the section of Lacey's regime channel to carry 10 m ³ /s of water in an alluvium of 0.50 mm.	CO2	06
Q. 4 Solve any TWO of the following		12
(A) Explain with neat sketch hydrological Cycle.	CO1	06
(B) Define hydrograph. Explain in detail components of hydrograph.	CO1	06
(C) There are four rain gauge stations existing in the catchment of a river. The average annual rainfall values at these stations are 780, 640, 400, 520 mm. Allowable percentage error in mean value of rainfall is 10%. Calculate optimum number of rain gauges in the catchment.	CO2	06
Q. 5 Solve any TWO of the following		12
(A) Describe the methods of controlling water logging.	CO1	06
(B) What do you mean by lift irrigation scheme? State its necessity.	CO3	06
(C) Explain with neat sketch rain water harvesting.	CO2	06

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